

SHIVAMSHU ROY

Buffalo, NY, USA | velamakannishivamshuroy@gmail.com

shivamshuroy.is-a.dev | linkedin.com/in/shivamshuroy | github.com/Shivamshuroy448 | medium.com/@shivamshuroy

EDUCATION

State University of New York at Buffalo

Master of Science in Engineering Science (Data Science)

Buffalo, NY

Aug 2026 – Expected December 2027

Bennett University

Bachelor of Technology in Computer Science (Data Science) CGPA: 8.93 / 10.0

Greater Noida, India

2020 – 2024

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, JavaScript, HTML/CSS

ML & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Time-Series (Holt-Winters, ARIMA), XGBoost, NLP, LLMs

Data & Cloud: Google BigQuery, PostgreSQL, MySQL, Firebase, FastAPI, Flask, Docker, ETL Pipelines

Analytics & Tools: Tableau, Power BI, Statistical Modeling, EDA, Git/GitHub, Linux/Bash, Jupyter

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

PROFESSIONAL EXPERIENCE

Data Scientist

Dec 2024 – Feb 2026

Nxtwave Disruptive Technologies

Hyderabad, India

- Architected end-to-end ETL data pipelines using SQL, BigQuery, and Python, automating cross-functional reporting across operational teams and reducing manual data processing latency by 40% across primary student databases.
- Designed interactive enterprise analytics dashboards querying 500K+ transactional student records; adopted across 4 major business departments, reducing cross-functional reporting cycles and operational turnaround by 35%.
- Built automated data validation and anomaly detection workflows using Google Apps Script and REST APIs, eliminating multi-team discrepancies and cutting executive report turnaround time from 2 days to near real-time.

Data Science Intern

May 2024 – Aug 2024

Tata Consultancy Services (TCS)

Hyderabad, India

- Developed Python-based automated Exploratory Data Analysis (EDA) pipelines and statistical verification scripts, standardizing data hygiene protocols and decreasing data cleaning and feature preparation time by 30%.
- Implemented scalable data transformation workflows and BI visualization reports in Tableau, enabling comprehensive real-time operational KPI tracking and performance monitoring across multi-disciplinary engineering teams.

PUBLICATIONS & RESEARCH

Stock Market Forecasting Using Advanced Time Series Analysis

[\[IEEE Xplore Link\]](#)

Published & Presented at the IEEE International Conference on Computer Vision and Machine Intelligence (CVMI-2023)

- Formulated optimized Holt-Winters exponential smoothing and adaptive moving average models for financial market index forecasting, achieving enhanced trend projection accuracy and superior fit over baseline architectures.
- Engineered specialized parameter optimization algorithms to minimize forecast error metrics (RMSE and MAE), significantly boosting forecasting stability and predictive precision across volatile historical financial markets.

PROJECTS

CheckmateLab – Browser-Based Chess Engine & Analytics

[\[GitHub\]](#) | [\[Live Demo\]](#)

- Engineered a client-side chess evaluation dashboard integrating Stockfish 16 NNUE via WebAssembly, delivering real-time positional depth evaluations, dynamic win probabilities, and opening book detection in the browser.
- Implemented dynamic move accuracy scoring algorithms and integrated Firebase OAuth authentication to calculate, analyze, and persist real-time player ELO ratings and game history progression across active user sessions.

ClearHire AI – Intelligent Application & Ghosting Risk Tracker

[\[GitHub\]](#)

- Developed an AI job application intelligence engine featuring Gmail OAuth synchronization to automatically parse recruiter correspondence, classify interview progression stages, and track application pipelines end-to-end.
- Formulated a predictive "Ghosting Risk" scoring algorithm analyzing company response velocity, status change frequency, and historical response patterns to trigger automated, context-aware follow-up email notifications.

Green Grid AI – Geospatial EV Charging Demand Prediction

[\[GitHub\]](#)

- Built machine learning regression models (Random Forest, Gradient Boosting) achieving an R^2 score of 0.88 to forecast municipal EV charging infrastructure demand and spatial utilization across 17,000+ spatial data records.